

Dynamic Adjustment of Monte Carlo Tree Search Simulations for A 9x9-Go-AI

This exposé will present a short overview of my bachelor thesis. The main topic of the thesis will be the evaluation of dynamic adjustments to the number of simulations executed by the Monte Carlo tree search (MCTS).

Idea

The thesis restricts Go to a small board size of 9x9 as a bigger board requires more capable hardware as well as more time to train. The core of the paper will be to implement heuristic estimations for a model based on AlphaZero that will increase and decrease the number of simulations executed by MCTS. This aims to lower the overall computational intensity. Previous work has largely been done in making

Experiments

Metrics

The model will be evaluated in terms of

- final playing ability and
- number of simulations and calculations executed during MCTS.

All models will receive the same amount of training time.

The thesis will introduce multiple heuristics.

- Hard-coded functions
- Learning models: e.g., Adaptive regressive model

Overview

1. Introduction to the topic
 - Problem space
 - Motivation
2. Background knowledge / Previous work
 - Go AI, MCTS
 - Rules of Go
3. Methodology

- Comparison of simulation number & Playing strength

4. Experiments

5. Discussion

6. Conclusion

Environment

The development and training of the models will heavily rely on Google's framework OpenSpiel as it provides a working implementation of AlphaZero. As the model is implemented in C++, the implementation of the heuristics will also happen done in C++. Preferably, training will be done on the university's cluster because of the heavy computational requirements of the model.

Timeline

Week 2	Implementation of models
Week 6	Training of models
Week 8	Evaluation of models
Week 10	Completion of writing
Afterwards	Fixing up